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測試深度學習網路和人類視覺系統的線性對應性

Testing the Linearity in the Correspondence between dCNN and Human Visual System

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摘要

本研究探討深度卷積神經網路與人類視覺反應之間線性特性的對應關係，特別聚焦於線性特性的兩項基本性質：可加性（additivity）與齊次性（homogeneity）。作為探討人類視覺與人工深度神經網路之間的對應，本研究結合了經過預訓練的 vgg16 卷積神經網路模型，作為心理物理實驗中刺激影像合成的計算工具，搭配經典心理物理學實驗範式進行。

本研究利用卷積神經網路階層式的架構，其非線性複雜度會隨著層級加深而提高。我們聚焦於 vgg16 網路中兩個早期卷積層（*conv1_1* 與 *conv2_1*，即前兩個處理區塊的第一層），這些層被認為與人類初級視覺皮質的神經處理特性相關，但在非線性程度上有所差異，為檢驗視覺處理早期階段的知覺線性提供了理想的測試背景。

本研究包含兩項心理物理實驗。在這兩項實驗中，刺激圖像皆由通過 vgg16 模型萃取自然場景圖像的深度特徵後，進行主成分分析，再經過調變後，最後經由相同網路合成還原為影像。

第一項實驗探討齊次性原則，方法是以預定的變動因子對深度特徵空間中的主成分進行比例縮放。第二項實驗則探討可加性原則，使用相同的深度特徵操弄與合成程序，分別比較三種條件：僅沿著第一主成分軸調變、僅沿著第二主成分軸調變，以及兩者加總調變。兩項實驗皆採用二選一強迫選擇範式與 Psi 方法進行閾值估計，參與者須判斷哪一張測試圖像在知覺上與參考圖像更相近。

結果顯示人類視覺處理存在偏離線性的現象。在實驗一中，偵測閾值未能穩定地對應於主成分的解釋變異量排序，顯示統計上的特徵重要性並不必然預測其知覺顯著性。在實驗二中，結果與特徵整合的線性可加性相矛盾，合併調變的條件並未穩定地在所有受試者與層級中產生低於單一主成分調變的閾值，反而支持「勝者全得」（winner-takes-all）機制而非線性總和的處理方式。至於較深層（*conv2_1*）是否相較於淺層（*conv1_1*）展現更強的非線性效應，結果則未得到一致支持。

關鍵詞：深度卷積神經網路（dCNN）、深度特徵、人類視覺系統、非線性模型、影像合成

Abstract

This study investigates the linearity in the correspondence between deep convolutional neural networks (dCNN or CNN) and human responses, particularly the two fundamental properties of linearity, additivity and homogeneity. To bridge the gap between biological vision and artificial deep neural networks, this research integrates a pre-trained vgg16 convolutional neural network model, as a computational tool for synthesizing stimuli images in psychophysical experiments with classical psychophysics paradigm.

Leveraging the hierarchical architecture of CNNs, which exhibit increasing nonlinear complexity across layers, we concentrate on the *conv1_1* and *conv2_1*, or the first convolutional layers of the first two processing blocks, of vgg16. These early layers are known to exhibit relatively high correlations with neural properties in the early visual cortex, yet differ in their degrees of nonlinearity. This provides an ideal testbed for examining perceptual linearity at early stages of visual processing.

The study comprises two psychophysical experiments. In both, stimuli were generated by modifying well-controlled deep feature representations derived from natural scene images passed through the feedforward structure of the vgg16 model. These modified features were then synthesized back into images using the same network.

The first experiment explores the principle of homogeneity by scaling a target principal component of the deep feature space with a predefined modifying factor. The second experiment investigates the principle of additivity using the same feature modification and synthesis procedure. Here, we compared three conditions: modification along a single feature axis (first principal component), modification along a second independent axis (second principal component), and a combined condition where both axes were modified additively. Both experiments use a two-alternative forced choice (2AFC) paradigm and the Psi method for threshold estimation, participants judged which of two test images was perceptually closer to a reference image.

Results demonstrated deviations from linearity in human visual processing. In Experiment 1, detection thresholds did not correspond consistently with principal component rankings based on explained variance, indicating that statistical feature importance does not directly predict perceptual salience. In Experiment 2, findings contradicted linear additivity in feature integration, with combined modifications failing to consistently yield lower thresholds than individual principal component conditions across participants and layers, supporting a "winner-takes-all" mechanism rather than linear summation. The hypothesized stronger nonlinear effects in deeper layers (*conv2_1* vs. *conv1_1*) received inconsistent support.

Keywords: dCNN, deep feature, Human visual system, nonlinear model, image synthesis

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1. Introduction

1.1. Background on Human Vision Models

In everyday life, humans are constantly exposed to a vast array of visual stimuli, necessitating rapid and efficient processing by the visual system. Understanding how the brain interprets such stimuli, ranging from basic patterns to complex scenes, is a central aim of visual psychology; such a process of processing basic patterns underlies higher-level functions like object recognition, spatial reasoning, and scene understanding (Andrews, Watson, Rice & Hartley, 2015; Long, Yu & Konkle, 2018). Over the past half-century, research on the human visual system has advanced significantly, with a growing body of work exploring both linear and nonlinear mechanisms underlying visual perception.

A major breakthrough in understanding early vision came in the 1960s with the work of Hubel and Wiesel, who demonstrated that neurons in the primary visual cortex (V1) respond selectively to specific stimulus features, such as the orientation of edges, in both cats (Hubel & Wiesel, 1959) and monkeys (Hubel & Wiesel, 1968). Their experiments involved inserting microelectrodes into the primary visual cortex (V1) of anesthetized animals to record the responses of individual neurons to controlled visual stimuli, such as bars of light at different orientations and positions. By systematically

varying these stimuli, they found that some neurons responded maximally to specific orientations and spatial configurations, which led them to classify the neurons into simple, complex, and hypercomplex types based on their receptive field properties. Since then, research on the early ventral visual pathway has expanded considerably.

The standard model of V1, which characterizes neuronal responses as a combination of linear filtering followed by a nonlinear normalization stage, was formalized by Heeger (1992) and further developed by Carandini, Heeger, and Movshon (1997). However, psychophysical evidence for contrast gain control mechanisms underlying such models had been proposed earlier by Ross and Speed (1991), who showed that adaptation to contrast alters perceptual sensitivity in ways consistent with divisive normalization. The standard model is based on three key principles: linear filtering, nonlinear rectification, and normalization, which together explain fundamental response properties of V1 neurons. The first stage of the model involves linear filtering, in which neurons act as spatially selective filters, responding preferentially to specific orientations, spatial frequencies, and phases of the visual stimulus. The filtering process is often mathematically modeled using Gabor functions, which closely resemble the receptive field profiles of V1 neurons (Jones & Palmer, 1987). Following this initial filtering stage, the output undergoes nonlinear rectification, which transforms the linear filter output by applying a nonlinear function such as half wave rectification (setting negative

values to zero) or full wave rectification (taking the absolute value). Finally, divisive normalization regulates neural activity by normalizing responses based on the summed activity of surrounding neurons, which explains contrast gain control and other adaptive visual processes (Heeger, 1992; Carandini et al., 1997).

These foundational findings led to the broader insight that visual processing in the brain is organized hierarchically, where each stage of the visual system extracts progressively more abstract representations from visual input. This hierarchical organization builds upon early computations, such as those modeled in V1, and extends to mid and high level visual areas. For instance, while neurons in V1 respond to local features like orientation and spatial frequency, neurons in later areas such as V4 and the inferior temporal cortex (IT) are tuned to more complex stimuli, including shapes, object parts, and entire objects (Felleman & Van Essen, 1991).

To computationally capture this structure, Riesenhuber and Poggio (1999) introduced the Hierarchical Model and X (HMAX), a framework for object recognition. The HMAX model extends the principles of linear filtering and nonlinear pooling by organizing them into a series of alternating layers, *simple* (S) units and *complex* (C) units, that mimic the function of neurons along the ventral stream. S units perform template matching akin to Gabor-like filtering, while C units pool over local spatial regions to build invariance to position and scale. By stacking these layers, the model

increases both selectivity for complex features and robustness to image transformations, echoing the progression from low- to high-level processing observed in the brain. HMAX thus serves as a bridge between early visual models and computational accounts of high-level object recognition.

The idea that human visual perception involves nonlinear processes has long been supported by psychophysical evidence. In one of the earliest demonstrations, Nachmias (1974) measured contrast discrimination thresholds and found that sensitivity decreased when a pedestal contrast was added, suggesting a compressive nonlinearity in the contrast response function. Similarly, Klein and Stromeyer (1974) used masking paradigms to show that the visibility of a test grating was reduced by the presence of another grating at a nearby spatial frequency, supporting the idea that gain control mechanisms operate across spatial channels. Building on these findings, Legge and Foley (1980) measured contrast detection thresholds using a pedestal-masking design and proposed a quantitative model in which a linear filtering stage is followed by a nonlinear transducer function and divisive normalization, successfully accounting for both facilitation and suppression effects observed in human performance. Foley (1994) later refined this model by showing that the shape of the contrast response function varies across spatial frequencies and depends on both excitatory and inhibitory

components, providing strong evidence that divisive normalization plays a central role in visual contrast processing.

Expanding on this tradition, Graham and Sutter (1996, 1998) investigated nonlinear processes in texture perception using element-arrangement textures designed to isolate responses from different spatial frequency and orientation channels. By systematically manipulating parameters such as spatial frequency, contrast, and background luminance, they demonstrated that texture channels, especially complex channels tuned to higher-order patterns, exhibited stronger nonlinear behavior, including contrast gain control consistent with normalization. Their findings indicated that normalization processes extend beyond basic contrast sensitivity and are involved in more sophisticated perceptual operations like texture segregation.

While Graham's work highlighted important nonlinearities in intermediate vision, similar conclusions had already been reached in broader and more comprehensive models. For example, Thomas and Olzak (1996) introduced a second-order LNL (linear-nonlinear-linear) model to explain how interactions between spatial patterns could produce texture-selective responses, and Hess et al. (2000) provided evidence for context sensitive second order mechanisms, showing that human perception of texture and form is shaped by both local nonlinear filtering and long range interactions. These

studies collectively suggest that normalization and nonlinear integration are central organizing principles of visual processing across both early and higher visual stages

More recent research also has highlighted the nonlinear nature of human vision, challenging traditional linear models. Carandini and Heeger (2012) revised the normalization model, proposing a "flexible normalization model", which suggests that normalization is not fixed but rather adapts based on stimulus properties and task demands. Golden and his colleague (Golden, Vilankar, Wu, & Field, 2016) introduced the concept of hyperselectivity, describing how a neuron’s selectivity is influenced by nearby neurons. Using the curvature of isoresponse surfaces(contours in stimulus space along which neural responses remain constant), they characterized the nonlinear selectivity of early visual neurons, revealing where combinations of inputs interact multiplicatively or suppressively, rather than additively(Golden, Vilankar, Wu, & Field, 2016; Vilankar & Field, 2017).

Furthermore, Bertalmío and his colleagues (Bertalmío, Gomez-Villa, Martín, Vazquez-Corral, Kane & Malo, 2020) conducted psychophysical experiments where subjects rated the perceived differences between image pairs to measure nonlinear characteristics of receptive fields. They found that the neural field equations accurately predicted these nonlinear responses in human vision and demonstrated that these nonlinearities enhance

visual processing efficiency by reducing redundancy in natural images and increasing the independence of neural representations compared to linear models.

1.2. The Principles of Linearity in Vision

Although substantial evidence supports the nonlinear nature of the human visual system, linearity remains a foundational concept for understanding many aspects of early visual processing. The application of linear systems analysis to vision was pioneered by Campbell and Robson (1968), who demonstrated that the human visual system processes images in terms of their spatial frequency components. By using grating stimuli and contrast sensitivity measurements, they provided strong psychophysical evidence that early visual processing could be modeled as a set of linear spatial frequency filters, an approach that laid the groundwork for subsequent theories of spatial vision.

Linear models offer a mathematically tractable and conceptually elegant framework for analyzing how visual information is encoded and transmitted, especially at early stages of the visual pathway. This approach was further developed in the work of de Valois and de Valois (1988), who integrated psychophysical and neurophysiological findings to demonstrate that receptive fields in early visual areas, such as V1, behave similarly to linear spatial filters, particularly when responding to low contrast stimuli. While linear

approximations may not capture all aspects of neural behavior, they remain a powerful tool for modeling the initial stages of visual input processing, such as orientation and spatial frequency selectivity.

It is important to note, however, that linear descriptions of neural responses have limitations. As early as the 1960s, physiological recordings, such as those by Naka and Rushton (1966), revealed that retinal and cortical neurons exhibit nonlinear response properties, particularly in how they saturate and adapt to stimulus intensity. These observations led to the development of nonlinear models incorporating mechanisms such as gain control, rectification, and normalization. Thus, while linearity provides a useful approximation in certain conditions, especially in psychophysical tasks with well-controlled stimuli, a full account of visual processing requires integration with nonlinear mechanisms that govern adaptation, context sensitivity, and perceptual inference.

In sum, linear and nonlinear frameworks are not mutually exclusive but rather complementary: linearity offer insight into the filtering and encoding of visual input, while nonlinear models capture the complexities of neural dynamics and perception. A comprehensive understanding of vision must therefore incorporate both perspectives to accurately reflect the layered computations occurring across the visual hierarchy.

In mathematics and systems theory, a linear system is defined as a system whose output

is directly determined by a linear transformation of its input. Specifically, a system is considered linear if it satisfies two key properties: additivity and homogeneity. In this study, we specifically examine these two properties :

Additivity: A system satisfies the additivity property if the response to the sum of two inputs equals the sum of the individual responses. This can be expressed mathematically as:

$$L(x+t) = L(x) + L(t)$$

Where x and t are the inputs, and L represents the linear system. If input x produces response $L(x)$ and input t produces response $L(t)$, then the system's response to the combined input $x+t$ should be equal to $L(x+t)$.

Homogeneity: A system satisfies the homogeneity property if scaling the input by a constant factor results in a proportional scaling of the response. This can be described by:

$$L(mx) = mL(x)$$

where x is the input, m is a scalar, and L represents the linear system. This means that if input x is multiplied by a scalar m, the output should also be scaled by m.

These properties provide the foundation for designing psychophysical experiments aimed at investigating the extent to which the human visual system adheres to or deviates from linearity.

Finally, neural mechanisms underlying perception further complicate the study of linearity. While early visual areas such as V1 exhibit response properties that involve relatively simple nonlinearities, higher-level areas along the ventral stream incorporate increasingly complex, nonlinear transformations that integrate contextual and semantic information. This hierarchical processing structure challenges the assumption that visual perception follows strictly linear principles at all stages.

Experimental methodologies such as computational modeling, neuroimaging, and psychophysical paradigms provide valuable tools for dissecting the balance between linear and nonlinear processes in vision. The present study seeks to contribute to this understanding by experimentally probing the principles of additivity and homogeneity with controlled psychophysical tasks using nature scene images processing via deep convolutional neural network.

1.3. CNNs and their Relationship with Human Vision

The roots of convolutional neural networks (CNNs) can be traced to early biologically inspired models such as Fukushima's Neocognitron (1979), which introduced a multilayered structure of feature-detecting and pooling units to model pattern recognition in a way reminiscent of the visual cortex. In parallel, the development of connectionist models by Rumelhart and McClelland (1986) helped establish the

foundations for learning in layered networks through distributed representations and parallel processing.

In 1989, LeCun et al. (1989) developed one of the first practical implementations of a trainable CNN, applying backpropagation (Rumelhart, Hinton, & Williams, 1986; Werbos, 1974) to recognize handwritten digits with high accuracy. Like the primate visual system, CNNs process input through a cascade of increasingly complex feature representations, with early layers detecting local edge-like patterns and later layers responding to object-level structures. Over the following decades, CNNs have evolved into the dominant architecture for visual recognition, achieving state-of-the-art performance across a wide range of tasks, from object classification to scene segmentation, by automatically learning hierarchical features from data (Krizhevsky, Sutskever, & Hinton, 2012; He, Zhang, Ren, & Sun, 2016).

Structurally, convolutional neural networks (CNNs) are built upon several key architectural elements that enable hierarchical visual processing. The fundamental building blocks include convolutional filters that detect local spatial patterns through learned weight matrices, pooling operations that reduce spatial resolution while retaining key activations, and nonlinear activation functions that introduce computational flexibility. These elements work together within local receptive fields,

where each neuron processes information from a spatially limited region of the input, similar to the receptive field organization observed in biological vision systems.

The power of CNNs emerges from their hierarchical feature extraction architecture, where successive layers build increasingly complex representations from simpler components. At the initial layers, CNNs typically learn filters that respond to basic spatial patterns such as edges, corners, and textures. As network depth increases, these primitive features are combined into more complex patterns, enabling the detection of object parts, shapes, and eventually whole objects. This hierarchical organization mirrors the broad computational strategy of the visual cortex, where early areas (V1, V2) encode basic spatial features, intermediate areas (V4) process complex features like curvature and texture, and higher cortical regions (inferotemporal cortex) contribute to complex object recognition (DiCarlo, Zoccolan & Rust, 2012).

Zeiler and Fergus (2014) provided compelling empirical evidence for this hierarchical progression by developing a deconvolutional network approach to visualize what CNN layers actually learn. Their method involved projecting feature activations back to the input pixel space through a series of operations: unpooling (reconstructing spatial locations using recorded pooling switch variables), rectification (passing activations through ReLU functions), and filtering (using transposed versions of the learned filters). By applying this visualization technique to different layers of their network, they

demonstrated that early layers predominantly detect simple visual elements like edges and color blobs, while progressively deeper layers respond to increasingly complex patterns such as object parts and complete objects. This systematic visualization revealed a clear progression from low-level features to high level semantic representations, closely paralleling the hierarchy from V1 to inferotemporal cortex in primate vision.

Building upon the architectural correspondences between human visual system and CNN , multiple neuroimaging studies have provided direct empirical evidence for functional similarities between CNN representations and human visual processing. Eickenberg et al. (2017) recorded neural activity using fMRI while participants viewed natural images and found that feature representations from intermediate CNN layers significantly predicted voxel activation patterns in object-selective regions of the visual cortex. Complementing these findings, Khaligh-Razavi and Kriegeskorte (2014) employed representational similarity analysis to systematically compare deep network layers with human brain responses, revealing that deeper CNN layers showed progressively stronger correspondence with higher level visual areas, while early layers aligned most closely with primary visual cortex activity patterns. Cichy et al. (2016) used magnetoencephalography (MEG) to measure temporal dynamics of visual processing and found that features from early CNN layers correlated with neural activity

in V1 and V2 within the first 100 milliseconds of visual processing. Furthermore, recent fMRI studies have shown that hierarchical processing in deep neural networks can predict voxel activations, supporting the notion that CNNs approximate aspects of biological vision (Henderson, Tarr, & Wehbe, 2023; Wen et al., 2018). These investigations have demonstrated CNN-brain correspondence across both spatial and temporal dimensions (Cichy et al., 2016), though with noted limitations in capturing higher-level visual representations (Xu & Vaziri-Pashkam, 2021).

These convergent findings have established CNNs as valuable computational models for understanding biological vision. However, the selection of an appropriate CNN architecture requires careful consideration of both computational capabilities and structural simplicity. In our study, we employed vgg16, developed by Simonyan and Zisserman (2015), which offers an optimal balance between computational effectiveness and interpretable feature extraction. For example, Gatys, Ecker, and Bethge (2015) demonstrated vgg16's interpretable feature representations by developing a neural style transfer method that could separate and manipulate content and style representations within the network. Their approach involved extracting content representations from higher convolutional layers and style representations using Gram matrices that capture texture patterns across multiple layers.

There were studies employed vgg16 to investigate the human visual system. Chen and colleagues' research using vgg16 to analyze human texture discrimination, which discovered that different CNN layers correspond to distinct visual properties like contrast versus higher-level aesthetic judgments (Chen, Lin & Chubb, 2019).

Vgg16 consists of 13 convolutional layers organized into five sequential processing blocks, with each block containing two or three convolutional layers followed by max pooling operations (Figure 1). In vgg16, all convolutional layers employ small 3×3 filters and ReLU activation functions.

For the implementation, we utilize the ImageNet-1k pre-trained vgg16 model available in the PyTorch torchvision library, which incorporates batch normalization after each convolutional layer with normalized running mean and running variance parameters optimized during training on the ImageNet dataset.

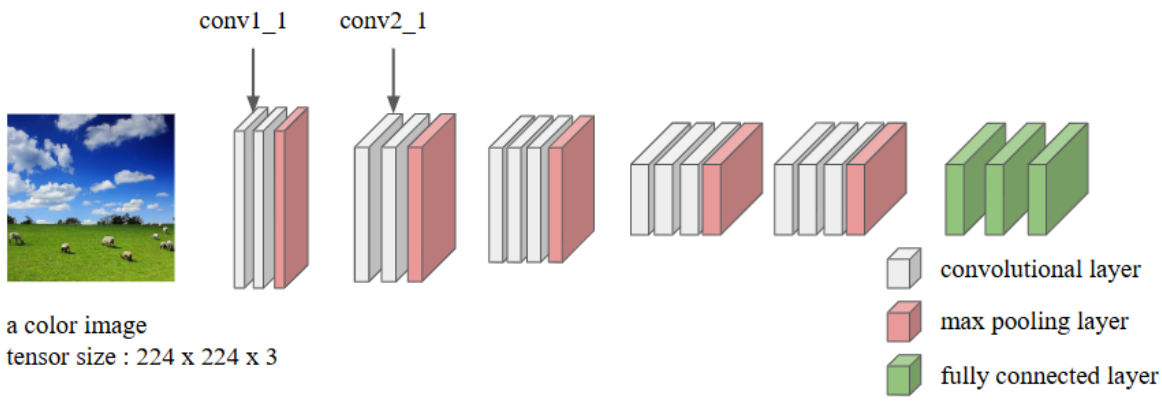

Figure 1. Structure of vgg16. vgg16 is composed of 13 convolutional layers, 5 max pooling layers, and 3 fully connected layers, totaling 21 layers. However, only 16 of these layers contain

learnable parameters, specifically the convolutional and fully connected layers. The network takes as input a fixed-size tensor of 224 × 224 × 3, typically representing a color image with three channels. The convolutional layers are systematically organized into five sequential blocks, containing 2, 2, 3, 3, and 3 convolutional layers, respectively. Each block is followed by a max pooling layer, which reduces the spatial dimensions of the feature maps. After the final convolutional block, the feature maps are flattened and passed through three fully connected layers, culminating in a softmax output for classification. vgg16 is widely used in both computer vision and cognitive modeling research due to its straightforward architecture and strong performance on image recognition tasks.

We focus specifically on features extracted from *conv1_1* and *conv2_1*, the first convolutional layers of the initial two processing blocks, to approximate computations during early stages of human visual processing. *Conv1_1*, with 64 filters of size 3×3, detects features directly from the raw input image. In contrast, *conv2_1*, with 128 filters of size 3×3, operates on the pooled output of the first block and captures more complex spatial patterns. This selection is grounded in empirical evidence demonstrating correspondence between early CNN layers and early visual cortex. As in Yamins and DiCarlo's study (2016), they conducted single-cell recordings in macaque inferotemporal cortex and revealed that early CNN layers capture low-level feature selectivity patterns closely resembling response properties of neurons in primary visual areas, including orientation selectivity and spatial frequency tuning. The temporal correspondence demonstrated by Cichy et al. (2016) provides additional evidence that

early CNN layers may capture computations functionally similar to early human visual processing.

Despite these compelling similarities, there are some fundamental differences between CNNs and biological vision systems. CNNs operate under artificially defined learning paradigms with discrete processing layers and feedforward information flow, whereas human observers develop through varied visual experiences with highly interconnected neural networks featuring extensive feedback connections and recurrent processing loops. CNNs learn through mathematical optimization, the backpropagation, while biological networks employ synaptic weight adjustments through incompletely understood mechanisms. These distinctions highlight challenges in directly equating deep networks with human vision while emphasizing their value as computational tools. While these parallels exist, it's important to note that CNNs were primarily developed as engineering solutions rather than as faithful neural models, and their processing differs substantially from biological vision in many respects (Lindsay, 2021).

Despite growing CNN integration in vision science research and accumulating evidence for functional correspondences between artificial and biological visual systems, there remains a significant gap in psychophysical validation of CNN-extracted features. Previous research has focused primarily on establishing parallels between CNN feature extraction and neural activity patterns measured through neuroimaging techniques

(Kriegeskorte, 2015; Yamins & DiCarlo, 2016). While these neurophysiological correspondences provide evidence for functional similarities, they do not directly address whether CNN-based representations adhere to established principles of human perceptual experience as measured through behavioral responses.

1.4. Incorporating CNNs in Psychophysical Experiments

The unique capability of deep CNNs to automatically extract meaningful hierarchical features without requiring explicit human-designed rules makes them particularly valuable for generating psychophysical research stimuli. Traditional stimulus generation approaches have relied on parametric manipulations of simple visual features, which are limited in capturing complex, multidimensional relationships characterizing natural visual environments. CNNs offer a transformative alternative by enabling the generation of stimuli that capture higher-order visual properties through sophisticated image synthesis techniques.

CNN-based image synthesis operates through iterative optimization processes guided by backpropagation. Given an initial image, usually a random noise, an objective function measures the similarity between the representation of the synthesized image and the representation of the target, such as content preservation or style transfer (Gatys et al., 2016). This technique enables systematic modification of images to test perceptual sensitivity to various visual properties (Wallis et al., 2017). Unlike traditional pixel-

based transformations, this approach allows control over high-level feature representations corresponding to meaningful perceptual dimensions, facilitating investigations into relationships between artificial network computations and human perception (Gatys et al., 2016; Dosovitskiy & Brox, 2016; Jagadeesh & Gardner, 2022).

In this study, we utilize CNN-based feature modification to investigate two principles of linearity in human vision. The concept of homogeneity suggests that by magnifying the effect of a feature, human perception would have a lower threshold on the component with more variance explained. While the concept of additivity suggests that the perceived effect of two combined visual features should be equal to the sum of their individual effects. To test these, we systematically manipulate CNN feature representations and generate stimuli that vary in controlled ways along different feature dimensions.

Using a pretrained vgg16 model, we extracted deep feature representations from images and modified specific feature maps to create synthesized stimuli. By analyzing participants' perceptual responses to these stimuli, we determined whether visual features interacted linearly or nonlinearly in perception. This approach provided a novel framework for studying the computational principles underlying human vision, leveraging the hierarchical feature extraction capabilities of CNNs by using nature scene images to probe fundamental psychophysical properties.

Despite powerful capabilities, utilizing CNNs for stimulus generation presents practical challenges, including computational cost and hardware limitations. Nevertheless, these are outweighed by significant advantages: CNN-generated stimuli can capture complex texture aligning with natural scene properties, increasing both perceptual complexity and ecological relevance compared to traditional parametric approaches (Yamins et al., 2014; Geirhos et al., 2019).

1.5. Research Gap and Aim of the Study

This limitation is particularly significant because neural activity patterns do not necessarily translate directly to perceptual experience or behavioral performance. CNN features may correlate with neural responses while failing to predict human perceptual judgments, or conversely, CNN manipulations may influence perception in ways not captured by neuroimaging measures. Therefore, direct psychophysical validation represents a critical step in establishing perceptual relevance of computational models.

Of particular importance is whether CNN based representations conform to fundamental psychophysical principles governing human visual perception, such as additivity and homogeneity in visual processing. These principles, extensively studied using simple parametric stimuli, represent basic assumptions about how the visual system combines and processes multiple information sources. Additivity refers to the principle that

perceived effects of combined visual features should equal the sum of their individual perceptual effects, while homogeneity suggests that magnifying a feature's effect should result in proportionally lower perceptual thresholds for detecting that feature.

This study aims to address these critical gaps by systematically evaluating whether CNN-extracted features conform to established psychophysical principles through carefully designed discrimination tasks.

By directly comparing CNN-based synthesized images using nature scene images with human psychophysical thresholds measured through behavioral discrimination tasks, this study addresses the critical gap in perceptual validation of CNN-based representations. This study not only bridges computational modeling with traditional psychophysical experimentation, advancing understanding of the intersection between CNN-based representations and human perceptual experience but also probes a new way to utilize nature scene images on the classical visual modeling paradigm.

2.Method

We investigated the linearity of correspondence between the human visual system and a convolutional neural network (CNN). The deeper layers in CNNs represent more complex and nonlinear feature transformations due to the cumulative effects of nonlinear operations such as ReLU activations and pooling. Therefore, we examined whether this hierarchical increase in representational complexity was reflected in human perception by comparing sensitivity to feature manipulations across layers. The thresholds were measured using the psi method.

To systematically manipulate feature content within each CNN layer, we applied principal component analysis (PCA) to the activation patterns of that layer. PCA is a statistical technique used to reduce the dimensionality of a dataset while preserving as much variance as possible. Given a set of high dimensional activation vectors, for example responses of a CNN layer to a nature scene image input, PCA identifies a new set of orthogonal axes, or principal components, that successively account for the largest possible variance in the data. The first principal component (PC1) captures the direction in activation space along which the variance was greatest; the second principal component (PC2) captured the greatest variance orthogonal to PC1, and so on.

In our study, we extracted the activations from a given CNN layer in response to the color images and computed the PCA basis from these activations. Each principal

component thus represented a common direction of variation in the feature space. To generate experimental stimuli, we projected the original image's activation onto individual principal components and then modified the image by exaggerating or suppressing variation along that component. This manipulation was performed independently for the top principal components (e.g., PC1, PC2, etc.) to assess their perceptual impact.

We hypothesized that modifications along PC1 would result in the most salient perceptual changes, followed by PC2, and so on. This hypothesis was motivated by the ordering property of PCA, which ensured that the early components explained the largest portion of variance in the activation space. However, we also acknowledged that variance alone did not guarantee perceptual relevance: some features contributing strongly to activation variance might not translate directly into noticeable perceptual differences. Therefore, part of our investigation was to empirically assess the relationship between statistical structure in the CNN activation space and human perceptual sensitivity.

Participants

Two participants took part in the first psychophysical experiment, while four participants, including the two participants who took part in the first experiment, were recruited in the second psychophysical experiment, all of whom participated in all

experimental conditions. They were naïve to the purpose and details of the study. Participants ranged in age from 20 to 29 years, with normal or corrected-to-normal vision and no history of visual disorders.

Apparatus

The experiment was conducted on an Apple Mac system. All experimental code was programmed in MATLAB 2020 with Psychotoolbox-3. The display monitor had a resolution of 1920 × 1080 pixels and with 60 Hz refresh rate. The monitor was calibrated with a PR655 spectroradiometer before the experiment to ensure consistent visual presentation. The test was conducted with participants seated 100 cm in front of the monitor in a dark environment.

Stimuli

The stimuli in this study were synthesized images created through a multi-stage process that systematically manipulated CNN feature representations. Involving extracting deep features from natural scene images, applying principal component analysis to identify key dimensions of variation, modifying these components in a controlled manner, and then reconstructing new images that embodied these specific feature manipulations (*Figure 3*).

Stimulus Selection and Preparation

We first manually selected 25 images from the Natural Scenes Dataset (NSD) (Allen et

al., 2022) based on the following criteria: (a) diversity of features : images contained various orientations, textures, and colors, ensuring strong representation in early visual processing stages, while early layers tended to filter those kinds of features (Güçlü & van Gerven, 2015); (b) void of salient objects : images lacked attention capturing symbols (e.g., red arrows, black-and-yellow warning signs) were chosen to minimize bias. We also avoided images containing human faces or prominent object centered compositions, in order to prevent potential confounding effects from face or object recognition mechanisms; (c) scene generalization : a mix of natural and urban scenes was chosen; and (d) perceptual complexity : images were chosen to be neither too simple nor too complex perceptually, ensuring that participants could detect subtle modifications without being overwhelmed.

Additionally, we verified that the feature distributions of selected images aligned with those of the larger NSD dataset to avoid statistical outliers. After preliminary testing, we found that including multiple different images in a single threshold estimation run could contaminate the results. This was because modifying principal components affected each image differently in terms of visual appearance, as images inherently possessed distinct features. As a result, we selected four specific images as the original images for image synthesis. This allowed us to analyze each image separately in the later analysis stage, thereby avoiding high variance across images and ensuring more reliable

threshold estimates.

Feature Generation and Modification

Challenges in Direct Feature Manipulation

In a preliminary experiment, we directly modified filter responses within target layers as independent variables in psychophysical experiments, and then selected the more important filters by applying PCA. However, this approach posed two major issues: (a) inconsistent scaling across layers : applying identical modifications to different filters led to disproportionate perceptual changes. Some filters produced noticeable differences, while others had minimal impact. Since a CNN was not designed to emphasize specific filters disproportionately, this discrepancy suggested that modifying raw feature maps directly did not accurately reflect their contribution to perception. (b) Interdependence of filters: a CNN relied on all filters collectively to extract meaningful representations. Assigning differential importance to specific filters was not justified, as all filters contributed to the learned representation. Thus, it was necessary to find a new way to control modification procedures if reducing the dimension of conditions was needed.

PCA-Based Feature Transformation

To address these issues, we developed an alternative approach involving feature normalization and PCA based transformation. The overall workflow of image stimuli

generation followed these steps: The input images were sampled from NSD at a resolution of 425 × 425 pixels and were downsampled to 224 × 224 using nearest neighbor interpolation to meet the input constraints of vgg16. The input images were fed forward through the pretrained vgg16 network to the target layers, *conv1_1* and *conv2_1* in our case, to obtain the output features. The output features were normalized using the running means and running variances from the ImageNet-1k pretrained vgg16. Principal component analysis (PCA) was then applied to the normalized features to transform them from feature space to PCA space, using PCA components trained on 10,000 images randomly sampled from NSD. The principal components of the features were modified by multiplying them by a scalar. The PCA and normalization processes were reversed, the modified principal components were transformed back from PCA space to feature space, followed by the inverse normalization. This ensured the feature data returned to the original space and scale, enabling image synthesis. Finally, the modified features served as the ground truth for image synthesis using the same pretrained vgg16.

The detailed procedure followed these specific steps:

Feature Extraction: We extracted feature representations by forwarding sampled images from the Natural Scenes Dataset (NSD) through the ImageNet-1k pretrained vgg16 model.

Normalization: We normalized feature maps using running means and variances obtained from the ImageNet-1k pretrained vgg16 model in the PyTorch torchvision library in Python. This normalization process followed the standard formula: $x' = (x - \mu) / \sigma^2$, where x was the feature value, μ and σ^2 were the running mean and variance of the feature values of the target layer, respectively. The running means and running variances were statistics accumulated during the training of the batch normalization layers, representing the expected mean and variance of each feature channel across the dataset. These values allowed for consistent normalization during inference, ensuring stable activation distributions without requiring batch-level computation.

PCA Projection: We projected the normalized features into PCA space to remove the redundancy (correlation between features).

Feature Modification: We scaled specific principal components with predetermined factors.

Reverse Transformation: We reverse-transformed the modified features back to the original feature space.

Denormalization: We denormalized the features and used them as ground truth representations to synthesize stimulus images via vgg16.

Feature Distribution Comparison

To ensure that our PCA training data aligned with the original ImageNet-1k dataset, we

compared feature distributions between the NSD images used for PCA training and the ImageNet dataset used in pretrained vgg16. Because retrieving histogram of full ImageNet was computationally infeasible, we instead extracted summary statistics, specifically, running means and variances, as representative measures of distribution similarity. We randomly sampled 10,000 images from the 73,000 available NSD images to compute feature histograms. This down sampling strategy was adopted to manage the high computational load associated with processing the full dataset.

Although both strategies were implemented to reduce computational demands, they were insufficient to resolve memory constraints. Consequently, we applied 2 × 2 average pooling during feature map extraction to further reduce memory usage while preserving spatial resolution.

Principal Component Analysis and Feature Projection

PCA was a widely used dimensionality reduction method that projected data into a new space where the first few principal components captured most of the variance. Given the large feature matrices and memory constraints, we employed Incremental PCA (Levy & Lindenbaum, 2000; Ross, Lim, Lin & Yang, 2008). Unlike standard PCA, which required the entire dataset to be loaded into memory to compute the full covariance matrix and perform eigendecomposition, IPCA operated in a memory-efficient manner by updating the principal components incrementally using mini batches of data. After an

initial batch was used to estimate the first set of components, the algorithm incorporated each subsequent batch by adjusting the internal estimates of the mean and principal axes based on the new data.

In our implementation, we used the IncrementalPCA module from scikit-learn, which performed updates using a partial singular value decomposition (SVD) on each incoming batch. SVD was a matrix factorization method that decomposed the data into orthogonal components capturing its internal structure. It expressed the data as the product of three matrices: one capturing how samples projected onto components, one diagonal matrix of singular values representing the variance explained, and one capturing the principal directions in feature space, which corresponded to PCA components. This approach allowed the algorithm to approximate the global principal components by gradually integrating statistical structure across batches, while also accommodating memory constraints that made it infeasible to process the full dataset all at once.

The feature tensors extracted from the target vgg16 layer had the shape (N, C, H, W), where N was the number of images, C was the number of filters (channels), and H × W were the spatial dimensions. To prepare the data for PCA, we reshaped each tensor into a two-dimensional matrix of shape (N × H × W, C) by flattening the spatial dimensions and stacking the activations across all images. This transformation treated each spatial

location in each image as an independent sample, with the filter responses (C) as the feature dimensions. The resulting 2D matrix was then used as input for Incremental PCA. The top five principal components from *conv1_1* explained 84% of variance, while the top ten components from *conv2_1* accounted for 66% of variance (*figure 2*). These components served as conditions in our psychophysical experiment.

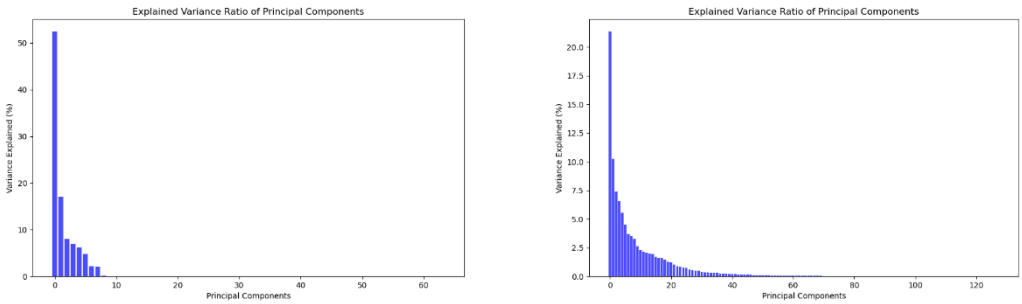

figure 2. Principal component analysis result. The explained variance of principal components in *conv1_1*(left) and *conv2_1*(right). The aim of the principal component analysis is to remove redundancy (correlation between features) and enable us to pick the more important PCs. The results exhibit a deep rather than flat shape, which may suggest that certain deep features can be discarded without significantly impairing performance. This implies that the CNN model can still achieve relatively high task performance even when some of its deep features are removed, highlighting a degree of redundancy or robustness in its feature representations.

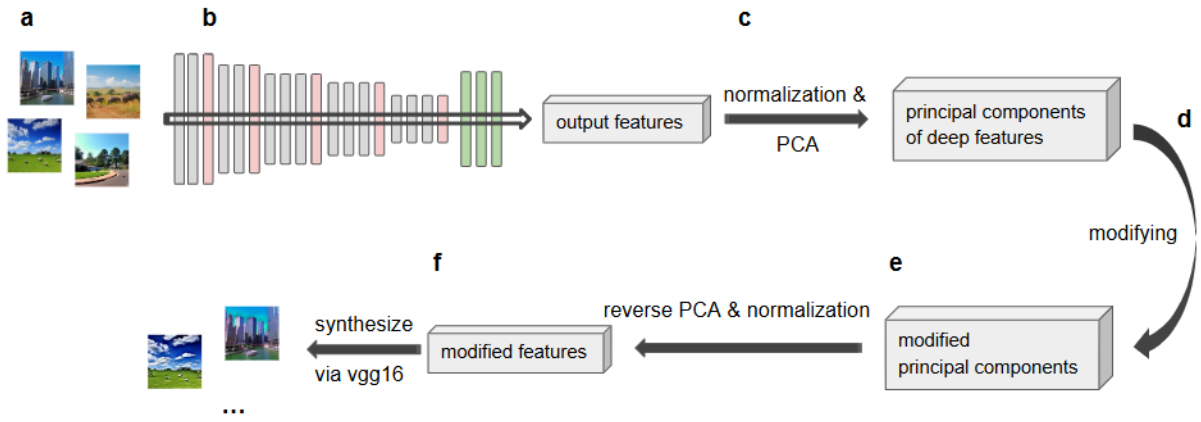

*Figure 3. T*he overall workflow of image stimuli generation. **a.** The input images were sampled from NSD at a resolution of 425 x 425 pixels. These images were downsampled to 224 x 224 using nearest-neighbor interpolation to meet the input constraints of vgg16. **b.** The input images were fed forward through the pretrained vgg16 network to the target layers, *conv1_1* and *conv2_1* in our case, to obtain the output features. **c.** The output features were normalized using the running means and running variances from the ImageNet-1k pretrained vgg16. Principal component analysis (PCA) was then applied to the normalized features to transform them from feature space to PCA space, using PCA components trained on 10,000 images randomly sampled from NSD. **d.** The principal components of the features were modified by multiplying them by a scalar. **e.** The PCA and normalization processes were reversed, the modified principal components were transformed back from PCA space to feature space, followed by the inverse normalization. This ensured the feature data returned to the original space and scale, enabling image synthesis. **f.** The modified features served as the ground truth for image synthesis using the same pretrained vgg16.

Feature Modification Protocols

To systematically manipulate feature representations, we applied modifications to the selected principal components to suit the requirement of each experiment:

Experiment I, Homogeneity: The top five PCs from *conv1_1* were selected, while the top ten PCs from *conv2_1* were selected. First, a selected principal component was multiplied by a scaling factor while keeping all other components constant. The scaling factors were logarithmically spaced to account for perceptual sensitivity differences. For *conv1_1*, modifications ranged from 1.1 to 10.869 in log scale across all five PCs. For *conv2_1*, modifications varied as described below: 1.1 to 10.869 for PC2 to PC6, 3.1 to 15.546 for PC1 and PC7, 5.1 to 26.083 for PC8 to PC10. The ranges were determined by several preliminary tests, in which we tested out where each PC's threshold was around and generated the image by that range. In total, 60 modification levels per principal component were created, yielding 900 feature variations per image for the psychophysical experiment.

Experiment II, Additivity: To investigate additivity, we prepared three experimental conditions: modification along the first principal component (PC1), modification along the second principal component, and simultaneous modification along both PC1 and PC2, referred to as the combined-feature modification condition. Both of the first two conditions shared the same set of stimuli as mentioned in the first experiment part. The

procedure for generating the stimuli in the combined-feature condition was the same as that of the homogeneity condition. The only difference lay in the feature manipulation step, where both PC1 and PC2 were simultaneously modified by the same scaling factor, from 3.1 to 15.546, after which the altered feature representation underwent the same synthesis pipeline to generate the synthesized image.

Stimulus Generation and Implementation Details

The modified feature representations were used to synthesize new stimuli images, following a similar approach to Deep Feature Interpolation (Upchurch et al., 2017). Unlike previous studies that directly interpolated features, our approach involved controlled modifications in PCA space before reconstructing images, as described earlier.

All image synthesis and feature processing were conducted using Python and PyTorch, leveraging the torchvision implementation of vgg16 with batch normalization (vgg16_bn). Feature extraction was performed via feedforward passes through selected layers of the network. In this model, batch normalization layers were inserted after each convolutional layer and before the ReLU. During inference, batch normalization normalized the output of each convolution using running estimates of the mean and

variance accumulated during training, followed by learned scale and shift parameters. This stabilized the feature distribution and could reduce internal covariate shift, potentially influencing the range and interpretability of activations used in our analysis. We used the pretrained weights provided by torchvision without fine-tuning or modifying the network.

The entire training and synthesis process was executed on an NVIDIA A2000 GPU with 32GB RAM in a CUDA environment.

Backpropagation in Image Synthesis

To reconstruct the synthesized stimuli images, we employed a backpropagation-based optimization method implemented in PyTorch. The goal of this process was to generate an image whose deep feature representation closely matched a predefined target feature. Specifically, by minimizing the discrepancy between the target feature (as described in the previous section) and the feature extracted from an input image, we could iteratively modify the image until its feature representation aligned with the desired deep feature.

The reconstruction process followed these steps:

1. Initialize an Image for Optimization: Instead of using a randomly initialized image, which often led to slow convergence and suboptimal reconstructions, we used the sampled original image from which the modified feature was derived.

This choice helped maintain structural coherence and accelerated the optimization process. We implemented 7000 iterations in *conv1_1*, 10000 iterations in *conv2_1*.

2. Forward Pass through vgg16: The initial image was passed through the pretrained vgg16 model up to the target layer. In our study, *conv1_1* and *conv2_1* after the batch normalization were chosen.

3. Feature Normalization and iPCA: The output feature map then went through the normalization process and PCA process aforementioned.

4. Feature Scaling and Reverse Back to Feature Space: We applied a modification factor to the chosen principal component. Then the matrix was transformed back from PCA space to feature space, and then reversed the normalization step right after. The modified feature was then treated as the ground truth feature.

5. Loss Computation: The difference between the forward passed feature map and the ground truth feature was quantified using the Mean Squared Error (MSE) loss function: $MSE = \Sigma_i(y_i - p_i)^2/n$, where y_i represented the ground truth feature values, p_i denoted the predicted feature values, n was the total number of observations, and Σ indicated summation over all elements.

6. Gradient-Based Image Optimization: We employed the ADAM optimizer in the backpropagation optimization (Kingma & Ba, 2014), an adaptive learning rate method that combined momentum (via first-order gradients) with element-wise learning rate scaling (via second-order gradient estimates). ADAM adjusted the learning rate for each parameter individually based on the estimated moments of past gradients, enabling faster and more stable convergence. In our setup, the optimization target was the input image itself. We defined a mean squared error (MSE) loss between the features of the current image and a reference feature vector extracted from the CNN. Gradients of this loss were computed with respect to the image pixels via backpropagation, and ADAM was used to iteratively update the image to minimize the loss. This process gradually refined the image until its extracted feature closely matched the predefined target feature.

This method effectively reconstructed stimuli images that preserved the feature modifications introduced in the PCA-based manipulation stage.

Final Stimulus Sets

In total, 3,600 images were prepared for the homogeneity experiment, encompassing 4 sampled original images, totally 15 PCs and 60 levels of modification. Specifically, in

this experiment, five principal components (PCs) were selected from *conv1_1*, and ten PCs were selected from *conv2_1*.

The selection of PCs as independent variables was guided by two primary considerations. First, only the top five PCs from *conv1_1* and the top ten PCs from *conv2_1* were included, as the explained variance beyond the cutoff principal component was too low. Modifications applied to lower-ranked PCs would not result in perceptually significant changes in the synthesized images. Second, incorporating all 64 PCs into the experiment would have substantially increased the complexity, requiring a larger sample of participants to ensure statistical power. Given the scope of this study, such an expansion was deemed unnecessary.

In the second experiment, we selected only PC1, PC2 and the combined-feature modification images for each of both layers. Totally 1,440 images were prepared for the additivity experiment : 4 sampled original images, three conditions for both layers and 60 levels of modification.

All the images used in both experiments were derived from the same 4 sampled natural scene images from the NSD.

Stimulus display

The actual stimulus display consisted of three images arranged on a neutral gray background, using an RGB pixel value of [128, 128, 128]. The three images were arranged as follows: The upper central image was the reference image. The type of the original images was randomly selected from the four sampled images. Two images were positioned below the reference image, one on the left and one on the right, aligned along the horizontal axis. One of these images was an unaltered copy of the reference image, while the other was a higher modified version based on the target feature manipulation (*Figure 4*). The Psi method was used to determine the optimal level of modification, ensuring efficient estimation of the perceptual threshold. The placement of the modified image was randomized across trials.

Procedure

A spatial two-alternative forced choice (2AFC) paradigm was employed in this study. To determine the perceptual threshold for each condition, we adopted the PSI method (Kontsevich & Tyler, 1999). While we acknowledged that the spatial 2AFC paradigm engaged peripheral vision and might yield different outcomes compared to a temporal 2AFC design (Peli, Garcia-Perez, Giorgi & Woods, 2004), the latter could introduce memory-related confounds due to the sequential presentation of stimuli. To minimize the influence of short-term memory and maintain a focus on immediate perceptual judgment, we chose the spatial version.

Each trial began with the presentation of a fixation point at the center of the screen against a neutral gray background, using an RGB pixel value [128, 128, 128]. This was immediately followed by an auditory cue, signaling that participants could press a key to begin the stimulus presentation at any time.

After the participant started the stimulus presentation, the stimulus display remained visible for 300 ms before being replaced by a neutral gray screen with a fixation dot at the center. Participants were then informed to indicate which of the two lower images appeared visually closer from the reference image by pressing a corresponding key on the keyboard. Immediately after responding, participants received auditory feedback which indicated whether their response was correct.

The stimulus duration was originally set at 200 ms, as previous studies had suggested that this duration was sufficient to minimize the likelihood of eye movements during stimulus exposure (Rayner et al., 1983). However, pilot testing revealed that 200 ms was insufficient for participants to reliably process complex, color-rich natural scene images. Given that color perception required additional processing time, particularly when assessing differences between similar images, the stimulus duration was extended to 300 ms to facilitate accurate perceptual judgments.

In the first experiment, a total of 60 experimental conditions were implemented, five PCs from *conv1_1* and ten PCs from *conv2_1*, four reference levels for each PC, as

described above. In the second experiment, a total of 24 experimental conditions were implemented, each of the first two PCs and the combined feature modification in both layers and four reference levels for each PC. Each condition was presented in a randomized order for each participant. Each experimental run consisted of 40 trials, with two additional practice trials at the beginning to familiarize participants with the task and minimize potential learning effects. Each of four image types were tested through each of the conditions. Each condition was tested in four repetitions in separate threshold measurements in experiment 2. In total, 240 runs were conducted for experiment 1 and 384 runs for experiment 2. Each took participants approximately 4 and 6.5 hours to complete, respectively.

Figure 4. The stimuli presentation. First, a fixation point showed up on the center of the screen. Then followed a three-image arrangement presentation presented for a 300 ms duration where the upper image is the reference image and one of the two images at the lower part of the screen is the same reference image, the other one is the image with higher degree of modification. Participants were asked to indicate which one was visually closer to the reference image using the keyboard.

3.Result

Stimulus Quality Validation

In the image synthesis session, we fixed the number of backpropagation steps rather than minimizing the final loss value. To ensure the validity of the synthesized stimuli, we employed two image similarity metrics, the L2 norm (Euclidean distance) and the Structural Similarity Index (SSIM). Both metrics assess the similarity between two digital images based on their pixel values. If the synthesized stimuli genuinely progress along a specific dimension (specifically principal components in our study), both metrics should reveal a gradient, ranging from minimal to substantial modification. *Figure 1* illustrates the validation results for our synthesized images across different CNN layers and modification levels. The heatmaps confirm that stimuli progressively shift along particular dimensions in pixel space. In addition, the absence of identical Euclidean distances or SSIM values across conditions validates that there are no two images being identical. Notably, both metrics showed consistent gradients in *conv1_1*, with the L2 norm revealing increasing pixel-level differences and SSIM capturing structural changes as modification strength increased. While in *conv2_1*,

Figure 1. Heatmaps of image similarity metrics comparing synthesized stimuli to the original images. **a.** Euclidean distance (L2 norm) to the original image. The leftmost column shows the original images sampled from NSD, and each column across the top corresponds to a target CNN layer. Within each heatmap, the horizontal axis represents the degree of modification,

increasing from left (minimal modification) to right (stronger modification). The vertical axis represents the principal components (PCs), arranged from top (higher-ranked PCs) to bottom (lower-ranked PCs). Lighter colors indicate lower Euclidean distances, meaning the modified image is more similar to the original. **b.** Structural Similarity Index (SSIM) between the modified and original images. As in **a**, the leftmost column displays the original images, and the top labels indicate the target layers. The horizontal axis within each heatmap shows increasing modification levels from left to right, while the vertical axis denotes principal components ranked from top to bottom. Darker colors represent higher SSIM values, indicating greater structural similarity to the original image.

Experiment 1: The Homogeneity Principle

Two participants, cih and cgw, took part in this experiment. Threshold data were analyzed separately for each participant and averaged across four sampled image types. To model the relationship between stimulus contrast and detection threshold, we applied Foley's model (Foley, 1994), which captures key nonlinear properties of human visual response. The model is defined as:

$$R = E^p / (I^q + Z)$$

Here, R denotes the observer's response, E is the excitatory input, and I is the inhibitory input, both rectified to non-negative values. The parameters p, q, and Z are the free parameters that define the system's nonlinearity and response baseline. In Foley's original formulation, E and I are products of contrast and sensitivity. This construct of the model itself makes it an appropriate choice for fitting detection threshold data in visual experiments involving varying stimulus intensities..

In this experiment, we used the model to fit threshold versus contrast (TvC) functions derived from feature modification stimuli. The goal was to examine whether the system's response exhibited a linear scaling property, formalized as $f(x \times a) = x \times f(a)$, where a represents the stimulus and $f(a)$ is the corresponding perceptual response. Under linearity, scaling the stimulus intensity by a factor of x should proportionally scale the response by the same factor x.

Conv1_1

For the *conv1_1* (*Figure 2*), TvC functions were measured for the top five ranked PCs. The results showed broadly similar threshold patterns across most PCs, with one notable exception. Both participants consistently showed elevated thresholds for PC4 relative to other PCs. The consistency of this anomaly across participants strengthens the evidence against complete homogeneity in early CNN layers. While PCs 1, 2, 3, and 5 showed comparable threshold functions, PC4's elevated thresholds suggest that not all principal components contribute equally to perceptual salience, even when they capture similar amounts of variance in the feature space.

Conv2_1

In the *conv2_1* (*Figure 3*), TvC functions were measured across ten PCs. Due to different contrast intensity levels used across PC conditions, threshold data were normalized using Weber fractions to enable direct comparison. As our data shows, thresholds appeared relatively uniform across all PCs, with no single component exhibiting notably higher or lower values. The intensity levels used were: PC1 and PC7: [1.0, 3.1, 6.139, 9.0]; PC2 to PC6: [1.0, 1.753, 2.903, 4.626]; PC8 to PC10: [1.0, 5.1, 9.906, 15.0]. Despite these different scaling factors, the normalized thresholds showed a remarkable consistency trend.

The contrast between *conv1_1* and *conv2_1* results provides insight into how feature

representation changes with processing depth. While *conv1_1* showed that PC4 was a consistent outlier, *conv2_1* demonstrated greater uniformity across components. This pattern suggests that deeper layers may develop more balanced feature representations, possibly due to increased abstraction and reduced sensitivity to low level image variations.

Principal Component Ranking and Perceptual Salience

Contrary to our initial hypothesis, principal components with higher explained variance did not consistently show lower detection thresholds. This unexpected finding challenges the assumption that variance in feature space directly translates to perceptual salience. The relationship between PC ranking and threshold sensitivity was more complex than anticipated, suggesting that perceptual relevance cannot be simply predicted from statistical measures of feature importance in CNN representations.

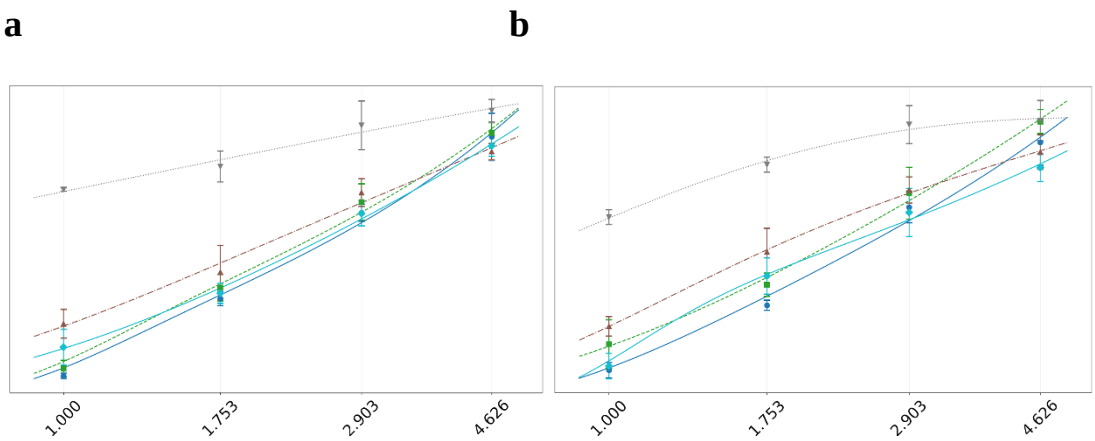

Figure 2. Threshold versus contrast (TvC) functions for five principal components (PCs) in the *conv1_1*. Each curve represents one principal component. Data points were averaged across four types of sampled original images. Curves were fitted using Foley's model: $R = E^p / (I^q + Z)$. The plots were generated using *semilogx* in pyplot library, with a logarithmic scale on the x-axis and a linear scale on the y-axis. The x-axis represents the raw modifier values, while the y-axis shows the threshold values in logarithmic units. **a.** TvC functions for participant *cih*. **b.** TvC functions for participant *cgw*.

a **b**

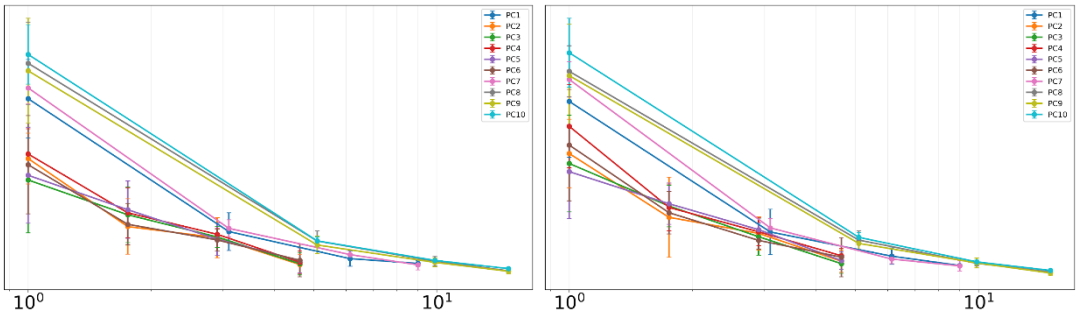

Figure 3. TvC (threshold versus contrast) functions for 10 principal components (PCs) in the *conv2_1*. Data points were averaged across four types of sampled natural scene images. To account for the fact that different PC conditions were tested using different intensity levels, the y-axis plots thresholds in terms of Weber fractions (threshold divided by base intensity). The intensity levels used for each PC condition were as follows: PC1 and PC7: [1.0, 3.1, 6.139, 9.0], PC2 to PC6: [1.0, 1.753, 2.903, 4.626], PC8 to PC10: [1.0, 5.1, 9.906, 15.0]. **a.** data for participant *cih*. **b.** data for participant *cgw*.

Experiment 2: Testing the Additivity Principle

Four participants, cih, cgw, lhc, and kys, participated in this experiment. Each condition was repeated four times. Detection thresholds were analyzed separately for each image type, with data points representing averages across four repetitions. As in Experiment 1, we fitted TvC data using Foley's model (Foley, 1994) to characterize perceptual responses under different feature manipulation conditions.

Experiment 2 evaluated the additivity principle by comparing detection thresholds

across three conditions: individual modification of PC1, individual modification of PC2, and combined modification of PC1+PC2. These conditions were tested across both CNN layers, *conv1_1* and *conv2_1*, for all participants. The additivity principle predicts that combined features should produce perceptual impacts greater than or equal to either feature alone, assuming linear integration. Thus, we expect that the combined modification condition should exhibit a lower threshold than the other two. The ultimate goal of this experiment was to test whether the system exhibits additive behavior, expressed formally as $f(a+b) = f(a) + f(b)$. In this formulation, $f(a)$ represents the perceptual response to a stimulus modified along one feature dimension, $f(b)$ represents the response to a stimulus modified along another dimension, and $f(a + b)$ represents the response to a stimulus modified simultaneously along both dimensions.

Figure 4 presents the TvC functions for all participants and stimulus conditions. Overall, threshold data were relatively consistent across participants and image types. Although we expected the combined modification condition (PC1 + PC2) to yield lower thresholds, based on the assumption that modifying two components simultaneously would enhance detectability. The results revealed that, for most participants and both CNN layers, the threshold curves for the combined condition were similar to those of the individual PC1 and PC2 conditions. This outcome suggests a potential "winner takes all" nonlinearity, in which the most salient feature dominates perceptual detection rather

than a linear integration of both components.

Interestingly, in the *conv2_1* for one of the four image types, participant data revealed even lower thresholds for the PC2 condition than for the combined PC1+PC2 condition. This unexpected result, limited to a single image, suggests that image-specific feature interactions or structural properties may influence perceptual salience under certain conditions.

When comparing across layers, the TvC curves for *conv1_1* displayed a monotonically increasing trend. In contrast, the *conv2_1* exhibited a dip-like shape in the TvC functions. This divergence between layers may reflect differences in the representational properties or nonlinear pooling mechanisms across the CNN hierarchy.

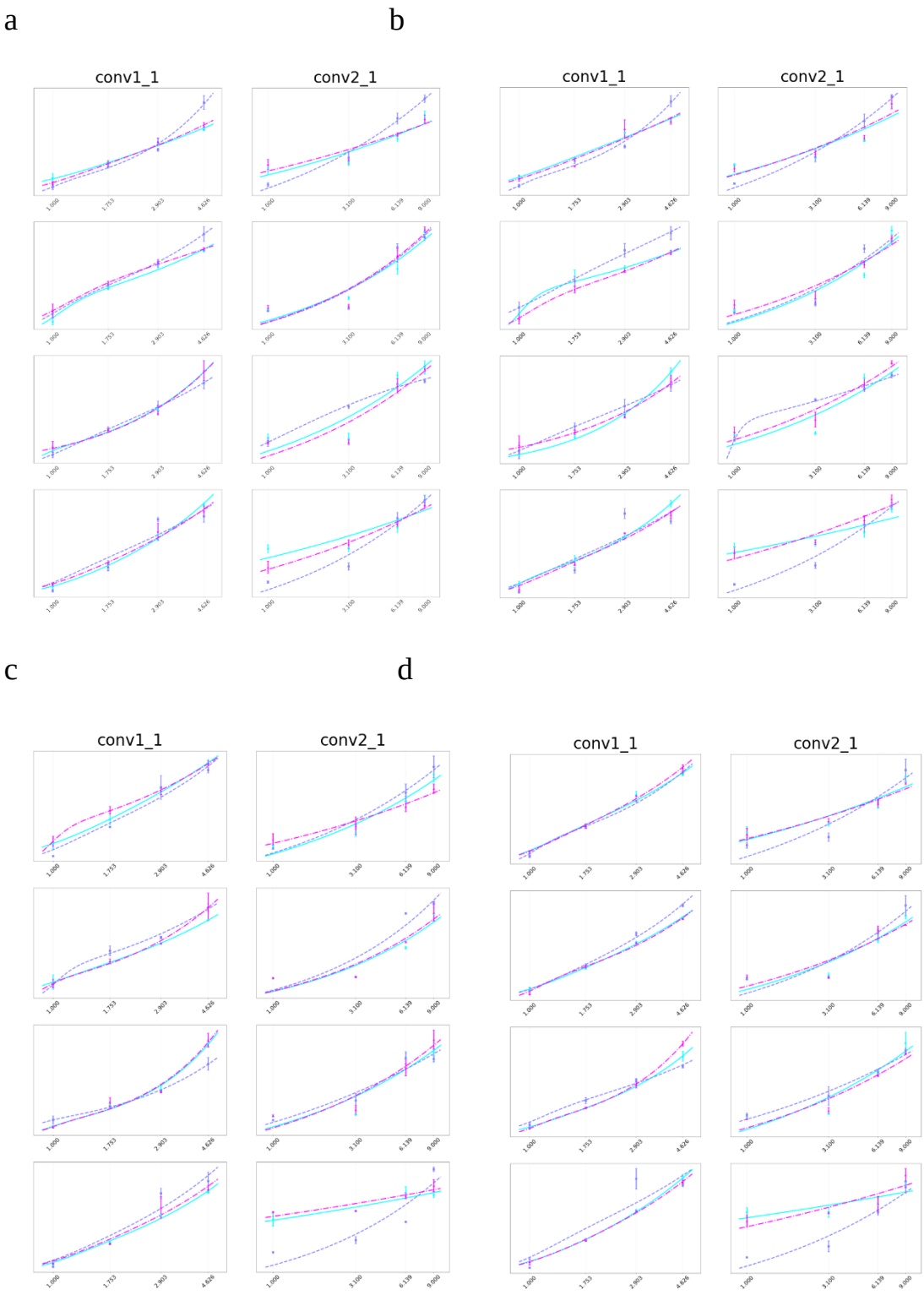

Figure 4. TvC (threshold versus contrast) functions from Experiment 2. The leftmost column displays the original images, and each column across the top corresponds to a target CNN layer. Within each plot, three curves represent the PC1, PC2, and combined modification conditions,

shown in light blue, purple and magenta, respectively. Curves were fitted using Foley's model: $R = E^p / (I^q + Z)$. Plots were generated using *semilogx* in pyplot library, with a logarithmic scale on the x-axis and a linear scale on the y-axis. The x-axis represents the raw modifier values, defined by the modification process in PC space described earlier. The y-axis represents threshold values in logarithmic units. **a.** TvC functions for participant *cih*. **b.** TvC functions for participant *cgw*. **c.** TvC functions for participant *lhc*. **d.** TvC functions for participant *kys*.

4.Discussion

This study introduced a novel methodology that incorporates CNN-synthesized natural scene images into classical psychophysical paradigms to investigate linearity principles in human vision compared to deep convolutional neural networks. Two main experiments tested the homogeneity and additivity principles using feature modified natural images generated through principal component manipulation of CNN representations.

In Experiment 1, we observed that detection thresholds did not align consistently with principal component rankings based on explained variance, challenging assumptions about the relationship between statistical feature importance and perceptual salience.

In Experiment 2, we did find evidence of nonlinearity in feature integration. Notably, across participants and layers, the combined modification condition did not consistently produce lower thresholds than the individual PC conditions, suggesting a “winner-takes-all” nonlinearity rather than linear summation. While our results confirmed the presence of nonlinearity in feature integration, we further hypothesized that nonlinear effects would be more pronounced in deeper layers of the network. Specifically, we expected to observe stronger nonlinear effects in *conv2_1* compared to *conv1_1*, based on the hierarchical nature of CNNs. While this trend was expected, the results were inconsistent and did not fully support our hypothesis. These inconsistencies may stem

from the specific details of our stimulus generation process or the complex properties of CNN representations. We will discuss these factors in more detail in the following sections.

CNN Computation and Human Vision

In this study, we employed Foley's model to fit psychophysical data and establish a comparison with convolutional neural network architectures. *Figure 5* presents a schematic comparison between Foley's model (Foley, 1994) and vgg16 processing structure. Foley's model describes how human visual responses to contrast are shaped by a balance of excitatory drive and divisive inhibition. This formulation reflects key nonlinear behaviors observed in early vision, such as contrast gain control and saturation.

Interestingly, this structure bears resemblance to the way convolutional neural networks (CNNs), such as vgg16, process visual input. In CNNs, the input is first passed through linear convolutional filters, followed by rectification through ReLU activation functions, which eliminate negative values. Afterward, a nonlinear pooling operation (max pooling in vgg16's case) aggregates spatial information and introduces invariance. While CNNs were not explicitly designed to model biological visual mechanisms, this sequence of linear filtering, rectification, and pooling loosely mirrors the excitatory-inhibitory structure of Foley's model, suggesting a potential functional analogy between

human and artificial visual system.

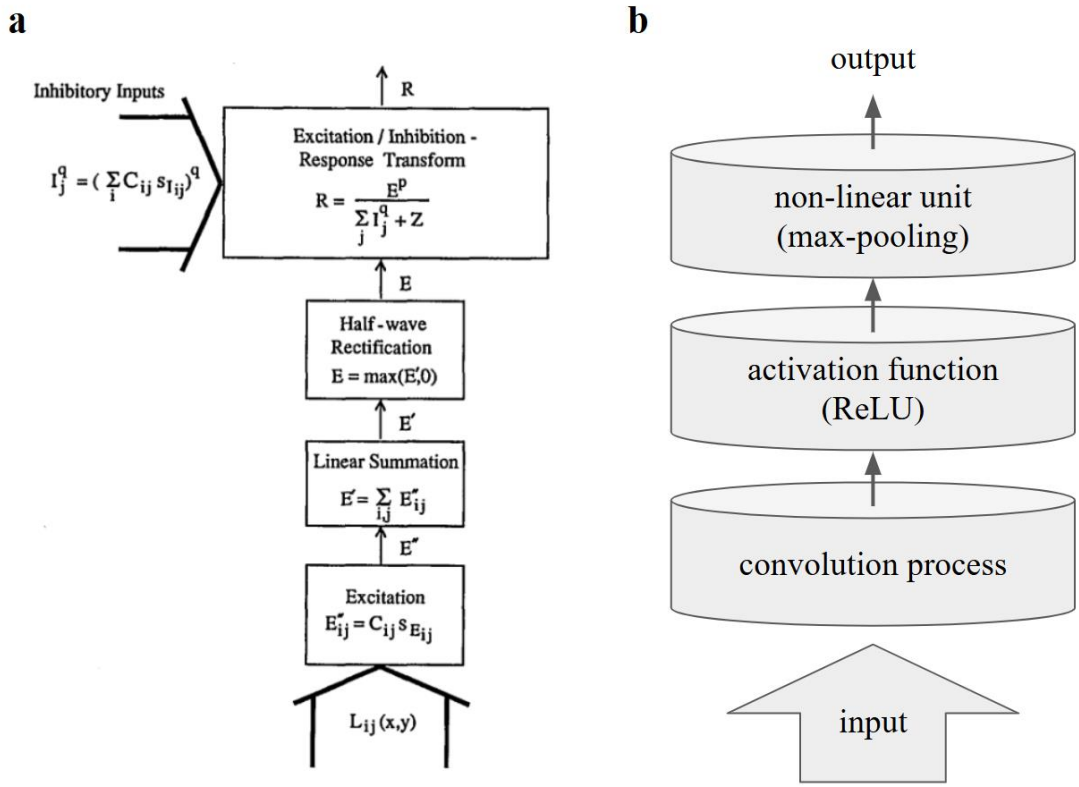

Figure 5. Comparison between Foley's model of human pattern-vision mechanisms and the structure of a convolutional neural network (CNN). **a.** The diagram reproduces Foley's model (1994), which models human visual response as a sequence of linear excitation, rectification, divisive inhibition, and response transformation. This model captures nonlinear phenomena such as contrast gain control and response saturation in early vision. **b.** The chart illustrates a simplified processing pipeline inside a part of CNN structure, including linear convolution, rectification via the ReLU function, and a nonlinear unit such as max pooling. While CNNs were originally designed for engineering purposes and not to mimic biological mechanisms, the structural correspondence between these two systems highlights a potential functional analogy

in how they process and transform visual input.

Note. a. reproduced from *Human luminance pattern-vision mechanisms: Masking experiments require a new model* by Foley, J. M. (1994), Vision Research, 34(6), 779-788. Copyright 1994.

Experiment 1

The unexpected finding that higher-ranked principal components did not consistently produce lower detection thresholds challenges fundamental assumptions of linear systems theory applied to vision. Our results suggest a more complex relationship between statistical importance and perceptual salience.

This dissociation may reflect differences between the optimization objectives of CNNs (which is classification performance) and the evolutionary pressures shaping human vision (which is survival-relevant feature detection). While PCA identifies directions of maximal variance in CNN feature spaces, these directions may not align with perceptually meaningful dimensions for human observers. The consistent elevation of PC4 thresholds in *conv1_1*, despite its statistical ranking, exemplifies this misalignment. As we observed the anomaly in early layers, particularly the PC4, suggests that even shallow CNN representations contain dimensions with varying perceptual relevance. This finding has important implications for understanding how artificial and biological systems organize visual information. Rather than uniform feature spaces where all dimensions contribute equally, our results point toward hierarchical organization where

some features are more perceptually salient than others. While the more unified patterns observed in *conv2_1* may reflect a normalization process where initial heterogeneity is progressively balanced through deeper processing.

Experiment 2

The results provide compelling evidence against simple linear additivity in feature integration. Contrary to the hypothesis that combining PC1 and PC2 modifications would enhance detectability through linear summation, the threshold data revealed that combined conditions (PC1 + PC2) performed similarly to individual PC1 and PC2 conditions across most participants and CNN layers. This pattern strongly suggests a "winner takes all" nonlinearity, where the most salient feature component dominates perceptual detection rather than integrating additively with other components.

This finding challenges models that assume linear integration of visual features and instead supports nonlinear pooling mechanisms in human vision. The absence of threshold reduction in combined conditions indicates that when multiple feature modifications are present, the visual system appears to rely primarily on the most detectable component rather than summing their individual contributions.

Layer Specific Processing Differences

In this experiment, we observed divergent patterns between *conv1_1* and *conv2_1* layers. The monotonically increasing TvC curves in *conv1_1* contrast sharply with the

dip like shapes observed in *conv2_1*, suggesting that different layers of the CNN hierarchy correspond to distinct stages of human visual processing with varying nonlinear characteristics.

This layer dependent variation may reflect the transition from more basic feature detection mechanisms at early stages (*conv1_1*) to more complex, nonlinear integration processes at deeper levels (*conv2_1*). The dip like patterns in *conv2_1* could indicate more sophisticated normalization or inhibitory mechanisms that emerge at higher levels of visual processing.

The unexpected finding that PC2 alone yielded lower thresholds than the combined PC1+PC2 condition in one specific image type within the *conv2_1* layer might suggest the context dependent nature of feature integration. This result suggests that image-specific structural properties or feature interactions can significantly influence perceptual salience, indicating that the winner-takes-all mechanism may be modulated by the specific visual context.

Such context dependency underscores the complexity of visual processing and suggests that feature integration strategies may be dynamically adjusted based on the statistical properties of the visual input, rather than following a fixed computational rule across all stimulus conditions.

Methodological Contributions and Innovations

Our PCA based approach for feature manipulation offers several advantages over direct filter modification methods. By working in principal component space, we achieved orthogonal manipulation dimensions that reduce interdependency issues common in direct filter approaches. This method provides more interpretable and controlled stimulus generation, enabling systematic investigation of feature space structure. However, we encountered technical limitations, particularly "overexposure" artifacts when pixel values exceeded valid digital image ranges (0-255) after feature modification and image reconstruction. This issue was most pronounced for PC1 modifications in *conv1_1*. While this aligns with the hypothesis that higher-ranked principal components preserve more variance of the original data, which corresponds to features that strongly influence global image statistics, it becomes particularly challenging and problematic when it comes to stimulus presentation. In our study, we addressed the overexposure issue by excluding any synthesized images in which more than half of the pixel values were outliers, specifically, pixels that reached the clipping boundaries of 0 or 255. This ensured that the final stimulus set remained within a perceptually valid range. Future implementations could address this through adaptive normalization strategies or constrained optimization approaches.

Natural Scene Stimuli in Psychophysical Research

The integration of natural scene imagery into classical psychophysical paradigms represents a significant methodological advance. Unlike traditional parametric stimuli, such as gratings or noise patterns, natural scene images provide a more realistic approximation of the stimuli encountered in daily life while maintaining experimental control through CNN-based feature manipulation. This approach bridges the gap between laboratory vision research and real world visual processing.

However, this methodological advancement comes with certain practical constraints. The computational intensity of image synthesis, particularly for deeper layers, represents a current limitation. Synthesis times increase exponentially with layer depth, constraining our analysis to early CNN layers. Future hardware improvements and optimization techniques may enable investigation of deeper layers, potentially revealing additional insights about high-level feature processing.

Limitations and Future Directions

Several technical factors limited the scope and precision of our investigation. The image synthesis process did not always achieve full convergence, with training losses remaining above zero even after extensive optimization (7000 steps for *conv1_1*, 10,000 for *conv2_1*). This incomplete convergence may have introduced variability in stimulus quality, potentially affecting threshold measurements.

Another issue is that the limited number of sampled original images, four natural scene images in our study, may not have provided sufficient feature diversity for comprehensive generalization. Participants occasionally reported focusing on specific image features during trials, potentially biasing their perceptual judgments. Future studies should incorporate larger, more diverse image sets and implement controls to discourage feature-specific strategies. A more diverse image set may also enhance the generalizability of the results to the broad range of visual experiences encountered in daily life.

Future Research Opportunities

Extended Layer Analysis

One of the motivations for using the vgg16 architecture in this study was to accommodate computational constraints. While vgg16 remains a widely used and well understood model, it lacks the architectural innovations found in more recent networks such as ResNet, EfficientNet, or vision transformers. Future studies could explore these more advanced architectures to assess whether their internal representations produce different perceptual effects. In addition, such cross architecture applications could provide insights into how different models represent visual information and whether their internal representations align with human perceptual organization.

Similarly, our focus on early convolutional layers (*conv1_1* and *conv2_1*) was partly

due to hardware limitations. Synthesizing images through deeper layers (e.g., conv4_1 or conv5_1) requires significantly more computational time and resources due to the increasing complexity and dimensionality of the feature representations. However, with access to more powerful hardware, future researchers could extend the analysis to these deeper layers. Doing so would allow investigation into whether deeper features, often associated with more abstract, semantic content, yield different patterns of perceptual threshold modulation and nonlinearity. The choice of layer could also be tailored based on specific domain interests, such as object recognition, scene categorization, or texture analysis.

CNN to the Human Vision

While CNNs provide powerful tools for generating controlled visual stimuli, our results highlight important limitations in treating them as direct models of human vision. The misalignment between statistical feature importance and perceptual salience, along with the variable support for integration principles, suggests that CNN representations capture some but not all aspects of human visual organization.

This finding emphasizes the importance of empirical validation when using CNN-derived stimuli in vision research. Rather than assuming correspondence between artificial and biological systems, researchers should systematically test these relationships through carefully designed psychophysical experiments.

It is important to reiterate that CNNs, including vgg16 used in this study, were originally developed as engineering tools optimized for performance, not as biologically faithful models. Their internal computations and learning dynamics often diverge in fundamental ways from those observed in the human visual system (Lindsay, 2021). While CNNs offer powerful and practical approximations for generating and analyzing visual stimuli, they are ultimately black-box models, in which the specific role and meaning of internal parameters, such as weights or feature activations, are not always interpretable. This stands in contrast to classical mathematical models of vision, where each component is defined analytically and linked to known perceptual principles.

The growing interest in using CNNs to study human vision reflects a valuable trend in vision science. Yet, future research should continue to interrogate the assumptions embedded in these models and develop hybrid frameworks that integrate data-driven performance with theoretical interpretability. Ultimately, a deeper understanding of both artificial and biological vision systems will likely emerge not from treating CNNs as replacements for visual models, but from leveraging them as experimental tools to probe and test perceptual hypotheses under controlled conditions.

5.Conclusions

This study demonstrates the feasibility and value of incorporating CNN synthesized natural scene images into classical psychophysical paradigms. Our PCA based feature manipulation approach enables systematic investigation of linearity principles using nature scene stimuli while maintaining experimental control.

While our methodology represents a significant advance in bridging artificial and biological vision research, the results might suggest that CNNs should be treated as experimental tools rather than direct models of human vision. The framework established here provides a foundation for future investigations that could incorporate more sophisticated architectures, deeper layer analysis, and broader stimulus diversity. The integration of deep learning tools with classical psychophysical methods offers promising avenues for advancing our understanding of human visual processing mechanisms. This methodological approach demonstrates the potential for investigating complex visual phenomena while maintaining the experimental rigor necessary for meaningful scientific insights, provided that the limitations and assumptions of both approaches are carefully considered.

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Appendix A

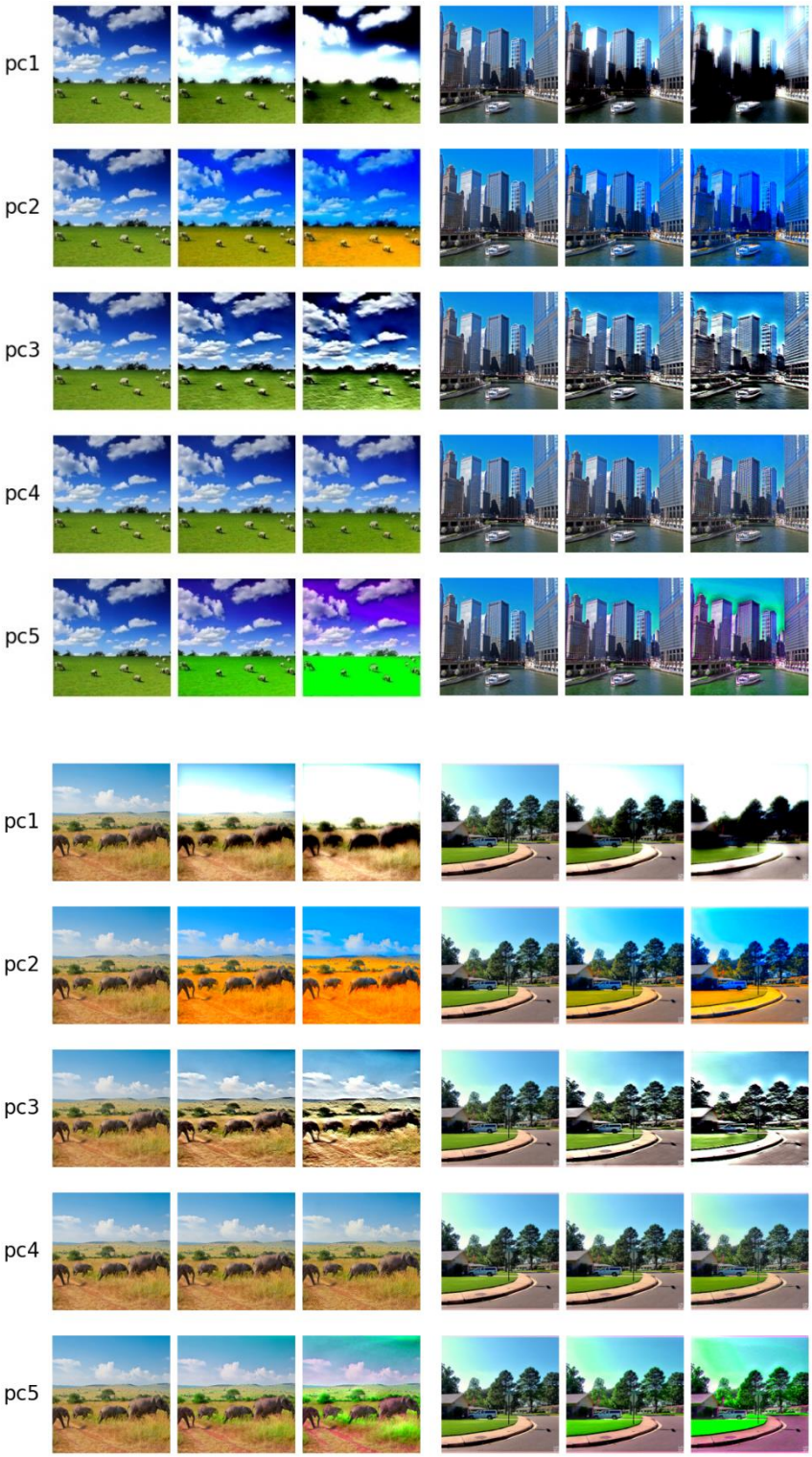

Figure A1. The synthesized image examples from modifying feature output of *conv1_1* using four images from NSD. From left to right indicate the degree of modifying. Each row corresponds to the different principal component modified before synthesis.

Figure A2. Examples of synthesized images generated by modifying feature outputs from the *conv2_1* using four natural scene images from the NSD. These stimuli were used in the 2AFC discrimination experiments. For each image type, the columns represent increasing degrees of feature modification from left to right, while each row corresponds to a different principal component modified before synthesis.

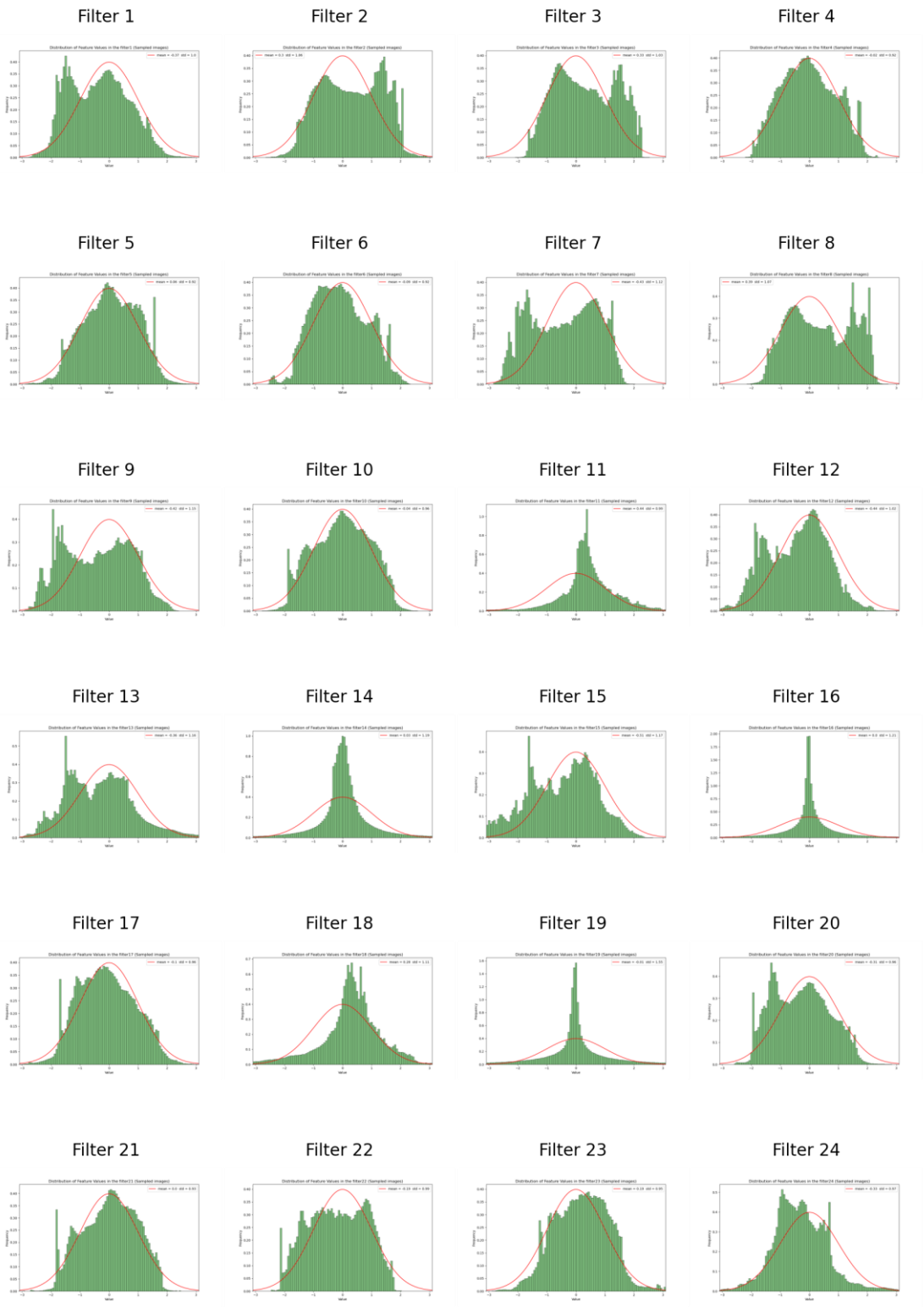

Figure A3. Feature value distributions of sampled NSD images compared to those of ImageNet-1k, with features retrieved immediately after the *conv1_1* layer. Green histograms represent the distribution of feature values from the sampled images, and red lines indicate the estimated ImageNet feature distribution. Due to hardware limitations, it was not feasible to store all ImageNet feature activations. Therefore, the red lines were approximated using the running

means and variances from the batch normalization layer following *conv1_1*, assuming a normal distribution. The purpose of this comparison is to ensure that the sampled images are not statistical outliers and that their feature activations are sufficiently similar to those of ImageNet images, thus validating their representativeness for analysis. The figure displays the first 24 filters out of 64 in *conv1_1*.

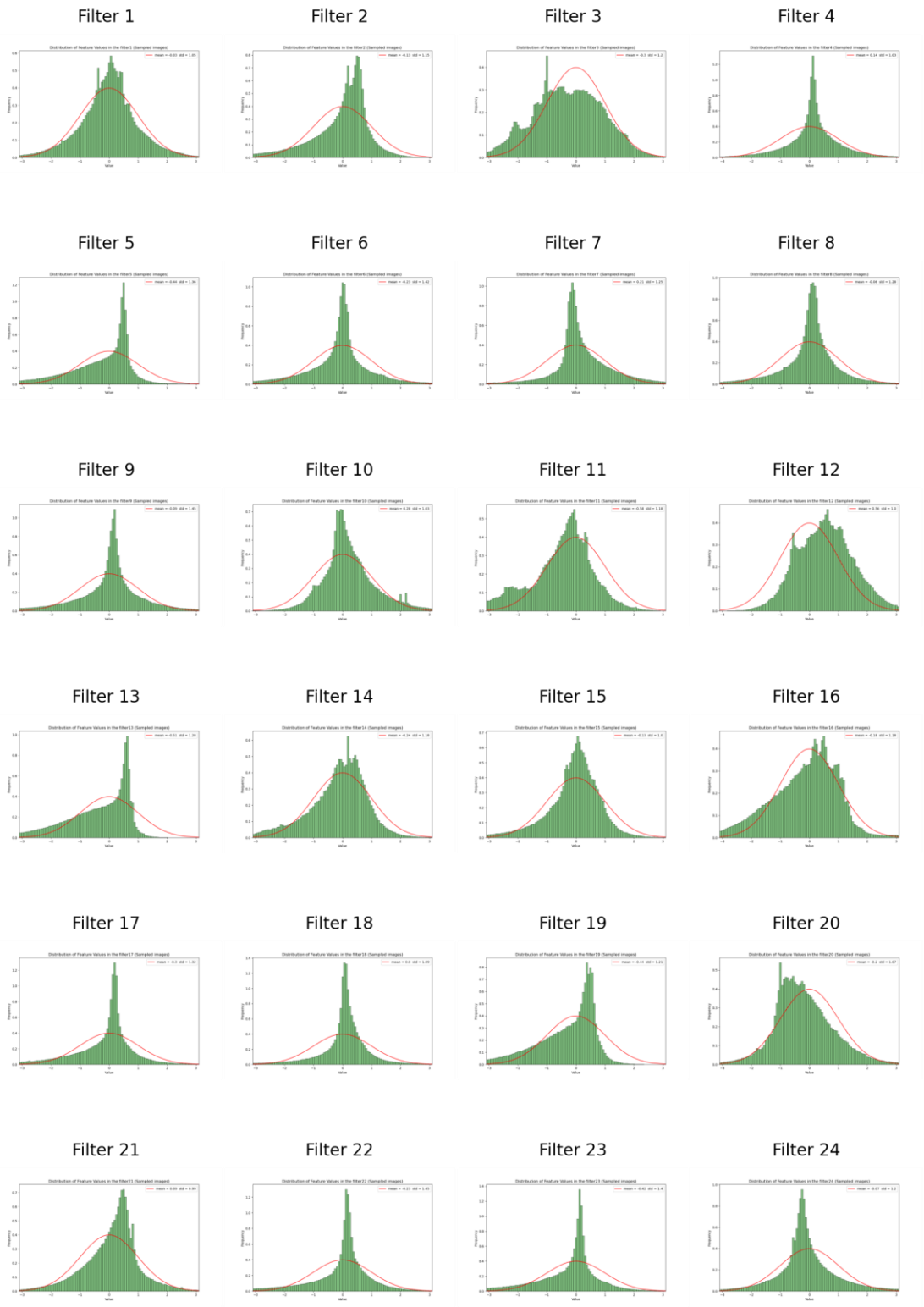

Figure A4. Feature value distributions of sampled NSD images compared to those of ImageNet-1k, with features retrieved immediately after the *conv2_1* layer. Green histograms represent the distribution of feature values from the sampled images, and red lines indicate the estimated ImageNet feature distribution. As in *conv1_1*, hardware constraints made it infeasible to store

all feature activations. The red lines were instead approximated using the running means and variances from the batch normalization layer following *conv2_1*, under the assumption of normality. This comparison was performed to confirm that the sampled NSD images do not substantially deviate from the distribution of ImageNet features, ensuring that they are appropriate representatives of the dataset. The figure displays the first 24 filters out of 128 in *conv2_1*.